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Serial participants of social media climate discussion as a community of practice: a longitudinal network analysis

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ABSTRACT

Serial participants of social media conversations are characterized by their sustained commitment to contentious politics. Previous research has mainly focused on the characteristics and motivations of individual serial participants. This study shifts attention to community level dynamics and applies the community of practice (CoP) theoretical framework to examine a group of serial participants who consistently tweeted about the issue of climate change over five years. Our findings reveal the patterns of structural evolution of the serial participant CoP, showing that a once loosely connected set of Twitter users have become more akin to a community of practice. With longitudinal network modeling, the study also reveals how status and hierarchies are established within this network. Our study also demonstrates the values of this CoP, showing that serial participants increased their connection with elites and diversify their connections through their participation.

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Introduction

Contemporary social-mediated movements, with leaderless and porous structures, have been credited for mobilizing publics at unprecedented speed and scales (Bennett, Segerberg, & Walker, 2014). Meanwhile, the low cost of involvement in social-mediated movements has also drawn criticism that many participants are only passively involved as ‘clicktivists,’ undertaking actions that are unlikely to lead to social change (Halupka, 2014; Lim, 2013). Questions remain regarding whether social-mediated movements are sustainable and the processes through which such movements can be sustained over time.

The identification of serial activists and the application of research on persistent activism to the Twitter context provides a missing piece that allows us to reevaluate the structures and actors that sustain social-mediated movements over time (Mercea & Bastos, 2016; Mercea, Karatas, & Bastos, 2018). Serial activists, as identified by Bastos and Mercea (2016), are Twitter users who tweet actively across multiple political protest events over time. They are defined by their magnitude of social media usage, geographic space (their activities often transcend national boundaries), and time. The most striking

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characteristic of serial activists is their sustained commitment to contentious politics. To understand this level of commitment and its impact, researchers need to look at the longitudinal aspect of social-mediated activism, and reevaluate the structure and evolution of a highly committed community and the value it offers to its participants.

Previous serial activist research has mainly focused on the characteristics and motivations of individual serial activists (Mercea & Bastos, 2016; Mercea et al., 2018). This approach has yielded important insights, but offers limited understanding of the community level aspect of serial participation. In this paper, we shift attention away from individual serial activists, and instead examine their communities. We first expand the concept of serial activists to *serial participants*, which refers to any individuals or organizations that contribute to social media conversations about the same issue across multiple issue-related events over time. Issue-related events create discursive opportunities to mobilize public attention to an issue and provide context for further public discourse, debate, and activism.

To understand serial participants as communities, we draw upon the community of practice (CoP) theoretical framework (Lave & Wenger, 1991; Wenger, MacDermott, & Snyder, 2002). CoP is a theoretical framework that describes how a group of members communicate, learn, and transform their practices through their interactions. The CoP framework positions individuals in a collective context and defines boundaries of communities through members' interaction patterns such as sustained tweets about an issue (Weiss & Domingo, 2010).

Guided by the CoP framework, we examine serial participants who consistently tweeted about the issue of climate change over five years. Our findings reveal the patterns of structural evolution of the serial participant CoP, showing that a once loosely connected set of Twitter users have become more akin to a community of practice over time. In addition, we demonstrate how status and hierarchies are established within this network: Organizational actors are preferred for interaction compared to individual Twitter users, suggesting that organizations continue to play a strong role in social media discussions of the climate issue. We demonstrate that by being members of the CoP, serial participants may increase their connection with elites and diversify their connections.

Serial participants and social-mediated movements

A line of recent research has identified and conceptualized 'serial activists' (Bastos & Mercea, 2016; Mercea & Bastos, 2016). These activists tweet profusely about multiple protests over extensive periods of time. Language or nationality barriers do not seem to discourage them as they rely on translation tools and peer networks to insert their involvement in any protest events that they see as relevant. Serial activist research departs from the line of research on 'social media influencers' is that serial activists are not necessarily elite social media users. Social media influencers are individuals who establish celebrity online (Duffy & Wissinger, 2017). Typical examples of influencers include the likes of movie stars, prominent bloggers, YouTube stars, 'Insta-celebrities' or other online personalities. In contrast, most serial activists are not politicians, celebrities, or media pundits (Bastos & Mercea, 2016). Instead, they are 'regular' people who dedicate time and effort to sustain communications about protests.

Longitudinal research reveals serial activists play crucial roles in stitching social movements together, ensuring sustained exposure to activist causes, aiding coordination of

actions, and ensuring the cascading distribution of activist messages (Mercea & Bastos, 2016). Serial activists differ from persistent activists, though the concepts are related through the shared element of persistence. The latter refers to activists who sustain commitment over time to specific activist causes whereas serial activists tend to take actions across geographic borders and engage in multiple activism events (Mercea et al., 2018).

Commitment and sustained momentum are important to all forms of social movements. Freelon, McIlwain, and Clark (2018) found that commitment (through repeated Twitter participation) predicted news media attention to Black Lives Matter. Similarly, Sommerfeldt and Yang (2017) argue that a movement that mobilizes a large number of participants but is short-lived may not contribute much to the actual realization of its mission(s). Profound social changes and significant policy changes often take decades and sustained efforts to realize (Sommerfeldt & Yang, 2017). Therefore, users' commitment considerably affects the effectiveness of both online and offline movement outcomes. In this regard, the research on serial activists helps to address crucial and yet little known questions such as: how do seemingly short-lived social media trends sustain and turn into long-lasting movements? How do social-mediated movements diffuse over time? What type of users drive such momentum, and how?

Previous studies have mainly focused on individual serial activists and their motivations, strategies, and behaviors. In this study, we broaden the scope of serial activists to examine serial participants (organizations as well as individual activists) in a contentious social issue over time. Our focal issue is climate change. The climate issue bridges multiple activist communities and spans many types of interested actors, including individual activists, advocacy groups, government actors, and media organizations. As a long-term social issue, attention to climate change is punctuated by a diverse set of issue-related events, including protests, government or NGO report releases, and natural disasters. Our analyses focus on the subset of climate issue actors that consistently participated in social media discussions of the issue over a long period of time, and in interaction with each other. Thus, the actors we study here persist in their Twitter communication behaviors across time, akin to persistent activists, but they also demonstrate the quite rare tendency to tweet across many different kinds of climate-related events, as serial participants (Mercea et al., 2018).

Our analyses go beyond a descriptive study of the network structure to explore factors that drive the formation and evolution of a community among these serial participants' in Twitter discussions of the climate issue. The shift in analytical focus from individuals to their communities helps to answer important questions, including: Can serial participants sustain communities over time and if so, how? Do serial participants' communities offer them functional values or relational resources? To guide our exploration in the direction of communities, we draw upon the community of practice framework, which is further discussed below.

Serial participants as community of practice

Community of practice (CoP)

A community of practice (CoP) is defined as 'a group of people who share a common set of problems, or a passion about a topic, and who deepen their knowledge and expertise in the

area by interacting on an ongoing basis' (Wenger et al., 2002, p. 5). There are three structural elements that are essential to the CoP framework: domain, community, and practice. Specifically, domain refers to the knowledge, topic, and issues around which a CoP is structured. A well-defined domain implies a shared commitment to the subject matter and will grow and evolve as the community develops. Community refers to the participants and their relationships through which information sharing and creation take place. Practice concerns their rituals, systems of meaning, and channels of communication. Practices are shaped and can evolve through members' mutual engagement, collective process of negotiation, and shared repertoires of everyday practice.

Although Lave and Wenger (1991) originally developed the concept of CoP to explain learning taking place in informal settings, the framework has been applied to a wide range of settings and topics. Studies have investigated how members of a CoP develop knowledge about carbon programs (Brown et al., 2016), develop innovative online journalism practices (Weiss & Domingo, 2010), and share routines and expectations for how to write blogs (Schmidt, 2007). In the current study, we apply the CoP framework to study a group of serial participants who consistently tweet across a diverse set of climate-related events over five years.

Social media can provide opportunities for geographically dispersed people to participate in CoPs (Gilbert, 2016; Komorowski, Huu, & Deligiannis, 2018; Stephansen & Coul-dry, 2014). Social-mediated CoPs are self-organizing, voluntary, and open participatory communities that are created and sustained through social media (Komorowski et al., 2018). Although they share commonalities with offline CoPs, social-mediated CoPs have unique features. Social-mediated communication is primarily text-based and may reduce social presence which hinders community building. Nonetheless, frequent interactions over time could mitigate such effect and this is why community building on platforms such as Twitter may require sustained, repeated interactions (Gilbert, 2016). Studies suggest that members of social-mediated CoPs use a range of tactics such as mentions, retweets, Twitter chat, and circles of storytelling to maintain interactions (Zappavigna, 2012). Technological features such as searchable hashtags and Twitter archives also change how members identify with each other. Archived conversation can function as web of interpretation to shape the construction of shared meaning (Zappavigna, 2012).

A CoP is not qualified based on forms of social structure or geographic boundaries. The size or the formation of the community is flexible. A CoP can consist of a few to hundreds of members. Members can be individuals, organizations, or a mix of both. A CoP is defined based on members' sustained commitment (Wenger et al., 2002), which is also a defining feature of serial participants. When we compare the definition of CoP and communities of serial participants, there are many similarities. For example, a CoP is voluntary and can best be characterized by members' consistent engagement in a domain-driven practice over time, as are communities of serial participants who Tweet about climate-related events. In addition, the existence of sustained interpersonal interaction is another key feature of CoP (Brown et al., 2016), which has also been documented in serial activist studies (Bastos & Mercea, 2016; Mercea & Bastos, 2016). Moreover, previous research also found that serial activists tend to coordinate their advocacies and support their peers through shared hashtags and framing strategies (Bastos & Mercea, 2016). Such strategies are similar to the concept of practice as outlined in the CoP framework (Lave & Wenger, 1991).

Nevertheless, there are also important differences between a typical CoP and a group of serial participants. For example, communications in typical CoPs are tailored to other members whereas communications in this serial participants' community can potentially be seen by the public. As such, it is unclear if sharing information in a Twitter CoP contributes to higher group cohesion. Previous studies of Twitter-based CoPs found sustained interactions contribute to group cohesion (Gilbert, 2016; Komorowski et al., 2018; Stephansen & Couldry, 2014). In this study, we also analyze if sustained interactions exist – and strengthen over time – among serial participants.

The CoP framework provides a useful theoretical framework to understand the values of sustained interactions among serial participants. We are interested in serial participants who participated in a number of climate change related events. Although our preliminary analysis suggests that they share domain knowledge and practices, it is unclear if they actually interact with each other and form a community. The first task therefore is to examine whether serial participants form a CoP, and whether their community can be characterized as loosely or tightly interconnected. To identify a serial participants' CoP, we look for evidence of repeated interactions among participants who can be classified as serial participants. In this study, we focus on the issue of climate change and participants who devoted sustained effort to discussing this issue on Twitter over a period of five years. Hence, we propose the following research question:

RQ1: Who are the participants involved in the climate change serial participant network over time, and do their interactions suggest the development of a CoP?

The structural dynamics of serial participants' CoP

The CoP framework maintains that a CoP is sustained by (1) a small group of core members who take on informal leadership functions or roles; (2) active members who participate regularly; and (3) periphery members who occasionally participate (Wenger et al., 2002). Although previous research recognizes the different roles of CoP members (Weiss & Domingo, 2010), the original framework does not explain how CoP members are positioned in different roles nor what mechanisms drive this process.

To extend the CoP framework, we draw insights from the network literature, which suggests that social selection is an important mechanism that brings communities together and produces core and periphery members in the process (Lewis, Gonzalez, & Kaufman, 2012). Social selection refers to the process through which social actors select others as social partners (Robins, Elliott, & Pattison, 2001). The mechanism is especially important in the digital space, where people have more choices when it comes to who to connect with and what contacts to keep.

Social selections can be influenced by both endogenous and exogenous factors. Endogenous network factors are structural features within a network that explain its structural tendency (Monge & Contractor, 2003). Variables are endogenous because they 'capture the extent to which relational properties of the network influence its self-organization' (Contractor, Wasserman, & Faust, 2006, p. 686). The network literature also suggests that network endogenous tendencies towards reciprocity, transitivity and the forming of three-cycles¹ are important evidence of social selections (Robins et al., 2001). The network tendencies contribute to relationship formations and the creation of clusters that hold

communities together. Although previous research has examined community dynamics in long-term friendship networks (Block, 2015; Rambaran, Dijkstra, Munniksma, & Cillesen, 2015), little is known if the serial participants' community is influenced by similar endogenous factors. Therefore we ask:

RQ2: Do network structural features (tendencies toward reciprocity, transitivity, and three-cycles) significantly influence the structure of the serial participants' community over time?

Additionally, social selections are influenced by exogenous network variables. Exogenous variables often refer to attributes of actors in networks that are determined by factors outside of these networks (Monge & Contractor, 2003) such as actors' status and resources. In other words, these are characteristics of networked actors that allow them to emerge as the preferred (or unpopular) social partners. In the case of serial participants, previous studies identified a wide range of actor types such as individual participants, media organizations, journalists, advocacy groups, advocacy media, etc. (Thorson, Edgerly, Kligler-Vilenchik, Xu, & Wang, 2016). Considering these actors possess different resources, it is likely that some types of participants are more popular than others in the social selection process.

Serial participants on Twitter may be individuals or organizations. Bennett et al. (2014) argue that in social mediated movements, 'participants engage with issues largely on individual terms by finding common ground in easy-to-personalize action frames' (p. 233). In other words, self-organized individual participants could operate in their ideal type without engaging and interacting with organizations. However, the literature on social movement organizations' (SMOs) strategic communication found that SMOs are important actors to monitor issues' evolution, carry out coordinated campaigns, garner public support and put pressure on corporations and governments (Obregón & Tufte, 2017). Although the literature is inconclusive on which actor, organization or individuals, is more important in the serial participant community, it is also likely that this status influences their desirability in communication networks. Therefore we ask:

RQ3: Do serial participants' actor attributes (organizational vs. individual accounts, type of participants) significantly influence the structure of the serial participants' CoP over time?

The value of serial participants' community of practice

Why do serial participants' communities matter? A CoP emerges in response to common interests or issues, and plays an important role in forming their members' participation in, and orientation to, the world around them (Caniglia, 2001; Wasko & Faraj, 2005). A CoP is central to many of its participants' identity construction, the forging of collaboration and coordination, and the establishment of norms and values.

Membership and network positions in communities of practice may powerfully shape participants' reputations among peers. For example, Wasko and Faraj (2005) examined why members of a legal association discussion forum help strangers. They found that active contributions help to enhance members' professional reputation among peers. In other words, higher status in the community of practice brings members respect, which may further influence how they are perceived by customers and general publics.

In addition, CoPs could also offer participants opportunities to connect with elites. This is because communities of practice, where many veteran participants and devoted

professionals interact, could potentially offer such an opportunity for serial participants to build elite connections and further elevate their status. Connections with elites can affect social movement repertoires, mobilization, and success rates (Caniglia, 2001; Yoo & Lee, 2009). In short, we analyze the value of the serial participants' network by considering whether central positions in the small circle of serial participants shape participants' opportunities to become visible in the general public network. We define this public network as the broader network of people and organizations discussing the climate issue on Twitter. The following research question is proposed:

RQ4: How do serial participants' network positions in their CoP affect their network positions in the larger Twitter conversation about climate?

Method

Data Collection

We employed the data mining service Sifter (Shulman, 2011) to capture historical tweets matching the keyword 'climate.' In order to track participation over time, we identified 12 events over a five-year period in which attention to the climate issue on Twitter was particularly high. For each event, we purchased the full firehose of tweets mentioning the keyword 'climate' during the particular 'spike' period ($N = 1,023,863$ tweets). We coded these events into five event types. From each event type, we selected the event with the largest number of tweets. These five events are the focus of our analysis (see Table 1 for a list of event types, events, duration, and the number of tweets during the event).

We then identified the group of more than 1000 Twitter accounts who tweeted in all five events. These are participants who showed sustained commitment on the issue of climate. However, to qualify as members of a CoP, these participants also need to interact with each other (Lave & Wenger, 1991). We defined interaction as mentions between participants because to mention someone, one needs to first be aware of the other contact. As such, unlike a retweet, mentions better reflect interactions based on preexisting relationships and experiences (Bastos, Raimundo, & Travitzki, 2013; González-Bailón, Wang, Rivero, Borge-Holthoefer, & Moreno, 2014). As such, we focused our analysis on users who interacted with another serial participant at least once during each of the five events. This step created five waves of repeated measures of 290 serial participants who interacted with others at all of the five different time points. These serial participants meet two defining features of CoP members: sustained engagement and interaction with other members (Lave & Wenger, 1991). The data on 290 serial participants were further coded and analyzed.

Table 1. Five online climate change spikes occurred between 2012 and 2015.

Time	Event types	Event	Duration	# of Tweets	# of Unique users*
1	Natural disaster	Hurricane Sandy	2012/10/30–11/1	171,678	105,144
2	Global conference	US China G20 climate announcement	2014/11/12–14	258,043	115,040
3	Religious event	Pope's encyclical on climate change	2015/6/17– 19	254,519	125,682
4	Activism event	People's Climate Movement	2015/10/13–14	178,766	102,302
5	Political-related news	President Obama's Facebook post on climate change	2015/11/9–10	160,857	83,486

Some users tweeted multiple times during one event but were only counted once for unique users.

Twitter coding measures

We conducted a content analysis of the Twitter users in the serial participant data set to identify characteristics of each account.

Individual attributes

Organization/individual account. This binary variable asks if a serial user is an individual or an organization (Krippendorff's $\alpha = .90$). Among the 290 serial activists, the coding identified more individual accounts (61.6%) than organizational ones (34.6%). See Appendix A for a full list of organizational serial activists.

Serial participant classification. In order to examine if the classification of serial participants affects their behaviors, coders coded the affiliation of serial users in terms of categories²: Mainstream media organizations, online-only media organizations, journalists, advocacy groups, advocacy media, political and governmental actors, celebrities, religious organizations, educators/scientists, environmental professionals, self-identified participants, and others (Krippendorff's $\alpha = .60$).

Network measures

We then calculated network measures for each serial participant as well as all for all users who participated in each climate-related event conversation during the events in our data-set. We used mention relations to calculate actors' network positions. On Twitter, mentioning requires the tagging of one account by another, which emphasizes the relationships between different accounts. Given that the interest of this study is on capturing how serial participants interact with one another, we focus exclusively on mention networks.

Network centrality. Centrality measures are some of the most important measures in network analysis (Friedkin, 1991). Centrality measures help to assess the level of importance that networked actors have in social networks (Freeman, Borgatti, & White, 1991; Marsden, 2002). We used four centrality measures to evaluate the network position and role of serial users. Degree centrality measures how often an actor communicates with others (including both actively mentioning others and being mentioned by others). Closeness centrality measures how close an individual is all other actors in a network (Freeman, 1978). Betweenness centrality refers to the number of times that an actor stands on the shortest path between nodes (Borgatti, 2005). Eigenvector centrality illustrates if an actor connects to well-connected others (Bonacich, 1972, 2007). Each measure evaluates network positions from different perspectives, emphasizing the number of connections, information circulation speed, brokerage/gatekeeper feature, and connections to others.

Network structural features. In this study, a number of endogenous and exogenous network features are modeled. Specifically, in terms of endogenous features, we modeled the three most important dynamics in long term relationship networks (Block, 2015): reciprocity, transitivity, and three-cycle. Reciprocation describes the likelihood of individuals to send ties to those from whom they receive a tie. Transitivity describes the tendency for two unconnected actors to be connected through a third party. Three-cycle describes a

triad of actors i, j , and h , among whom, a tie from i to k , and tie from j to h , and a tie from h to j is present.

Network configurations can also be affected by exogenous characteristics. In this study, we modeled if serial participants' status as an organization or individual or serial participant types influence their selection of contacts. Specifically, we modeled if alter or ego's status influence tie formation. In addition, we also model if homophily is present, where serial participants prefer others with similar characteristics.

Analysis

In order to reveal the serial users and understand factors that shape the evolution of their networks, as well as understand the value of their community, three types of analyses were used in the current study. The first is social network analysis. Social network analyses were performed to reveal the network positions of all participants' accounts and serial participants' accounts. Accounts' degree centrality scores were calculated using the sna package in R. Gephi was used to visualize the networks (Figure 1).

The second is stochastic actor-oriented model (SAOM) with the simulation investigation for empirical network analysis method implemented by RSiena (Simulation Investigation for Empirical Network Analysis) in R. We use this approach to model factors that significantly shaped structural changes in the serial participants' network. SAOM assumes that individual actors' decisions to create or terminate ties drive network change, and that network evolution is a product of actor-driven micro-mechanisms. RSiena allows for examination of the development of the network of serial users over time, while taking serial users' initial and changing network positions and account characteristics into account (Snijders, Lomi, & Torló, 2013).

The third analysis is mixed-effects regression models for longitudinal data. We use this method to assess if serial participants' network positions in the serial participant community significantly shaped their network positions in the general public network, controlling

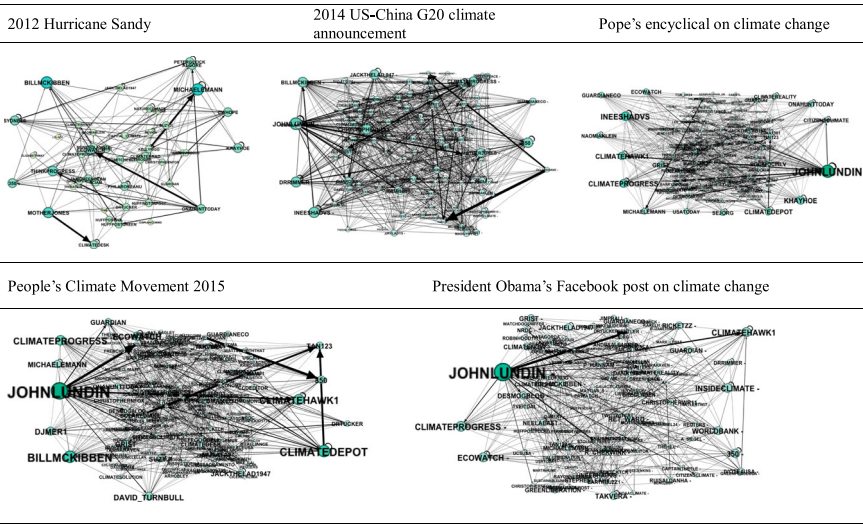


Figure 1. Serial participant mention networks in five activism events.

for other factors. As an extension of classic statistical procedures, linear mixed model (LMM) provides flexible analysis in correlated longitudinal data and allows researchers to model the covariance structures that represent random effects (Verbeke & Molenberghs, 2009). We utilized the lme4 Package in R for this part of the analysis.

Results

Network Descriptives

The five networks each contain (the same) 290 actors. According to the Rsienna manual, this is a reasonable size (Snijders, Van de Bunt, & Steglich, 2010) to model. A close look at the five waves suggests that the network densities stay relatively consistent with a tendency to decrease (period 1 = .008; period 2 = .012; period 3 = .008; period 4 = .006; and period 5 = .006). This means the CoP remained stable over five years with a slight tendency for actors to decrease interactions. Average degree tends to drop over time (period 1 = 2.169; period 2 = 3.410; period 3 = 2.248; period 4 = 1.741; and period 5 = 1.638) with the number of ties also dropping over time (period 1 = 629; period 2 = 989; period 3 = 652; period 4 = 505; and period 5 = 475). In addition, the analysis for change statistics suggests a gradually stabilizing process, such that distance between periods become smaller over time and Jaccard indexes remain relatively consistent (period 1→period 2, distance = 1358, Jaccard = .087; period 2→period 3, distance = 1311, Jaccard = .112; period 3→period 4, distance = 899, Jaccard = .125; period 4→period 5, distance = 776, Jaccard = .116).

RQ1 explores the characteristics of the most prominent serial participants involved in the climate change movement network. We identified the top 20 serial users with the highest weighted degree (sum of in- and out- degree) for each issue event in the Twitter data set. This measure indicates the overall attention a serial user received from members of the serial participant community, as well as how many times the user sought to bond with other serial users. Since there were five issue events in our dataset, this approach produced a list of 100 accounts. A close examination of user profiles suggests no obvious pattern of organizational and individual distribution across events of prominent weighted users. The only exception was the Pope's Encyclical on climate change event, in which 19 out of the 20 highest degree accounts were individual accounts.

Table 2 lists the types of top serial participants by event. Many of those serial participants remained highly central across events. In fact, out of the total of 100 top participants across five events, many accounts overlap from event to event and only 53 are unique. Among the 53 unique accounts, 33 were individuals and 20 were organizations. Overall,

Table 2. Top 20 (weighted degree) serial users categorization at five spike events.

Spike events	Organization vs. individual accounts	Top type of serial user accounts
2012 Hurricane Sandy	Individuals (7); Organizations (13)	Online only media (5) Advocacy groups (3)
2014 US China G20 climate announcement	Individuals (13); Organizations (7)	Individual activists (7)
Pope's encyclical on climate change	Individuals (19); Organizations (1)	Individual activists (12)
People's Climate Movement 2015	Individuals (14); Organizations (6)	Individual activists (5) Advocacy groups (3)
President Obama's Facebook post on climate change	Individuals (9); Organizations (11)	Advocacy groups (5) Individual activists (4)

the analysis revealed that the most active and prolific users tend to stay prominent across events. On average, the serial participants' number of followers increased 7.2 times between 2012 and 2015. In the meantime, the number of accounts these serial users follow only increased an average of 2.3 times, which may suggest that serial users are more cautious in terms of choosing whom they follow.

In order to address research question 2 and 3, Rsiena was applied to test the effect of structural and actor-level factors on the network evolution of serial participants' community. Wave1 was dropped from the analysis because when we included all five waves, the Jaccard index (which captures the changes in the network) is smaller than .03 due to insufficient change between wave 1 and wave 2. When Jaccard index is too small, the data cannot support the modeling procedure (the threshold suggested by Snijders et al., 2010). Once we removed wave 1, we computed Jaccard index again, with a value larger than .07, this time the model converged, with a converge ratio of .15.

RQ2 asks if network structural features (tendencies toward reciprocity, transitivity, and brokerage) significantly influence the structure of the serial participants' CoP over time. Results of the RSiena analyses are presented in Table 3. Specifically, the negative outdegree effects showed that serial participants are very selective in their interaction with other serial participants. When they do build connections, the positive and significant reciprocity effect suggests that they tend to reciprocate ties received.

The positive transitive triplets effect indicated a tendency towards transitive relationships. This result, paired with the negative and significant three-cycles effect (which suggests that serial participants are significantly less likely to interact with contacts of their own contacts), reveal the existence of local hierarchies, with some serial participants receiving more interactions than others within the same triplet. The result is consistent with previous longitudinal network studies about teen friendship formation (Rambaran et al., 2015), and suggests that over time some serial participants will emerge as very popular within the community.

Table 3. Results of the SIENA multivariate network analysis: estimates of between-network effects for friendship and antipathy controlled for structural network effects, and individual and behavioral dispositions.

	Estimate	Standard error	Convergence <i>t</i> -Ratio	<i>p</i> *
<i>Rate constant</i>				
Serial activist network rate (period 1)	19.48	(1.77)	−.11	**
Serial Activist network rate (period 2)	10.69	(.86)	.01	**
Serial Activist network rate (period 3)	8.30	(.77)	−.05	**
<i>Structural network effects</i>				
Outdegree (density)	−2.95	(.03)	−.10	**
Reciprocity	1.27	(.11)	−.01	**
Transitive triplets	.68	(.03)	−.08	**
Three-cycles	−2.23	(.62)	−.08	**
<i>Serial activist characteristics</i>				
Organization/individual alter	.37	(.03)	.02	**
Organization/individual ego	−.33	(.05)	.01	**
Organization/individual similarity	.04	(.05)	−.08	n.s.
Actor type alter	.05	(.03)	.06	n.s.
Actor type ego	.07	(.04)	.02	n.s.
Actor type similarity	−.3479	(.06)	−.09	**

*The significance of these evaluation function weights was assessed by dividing the estimates by their standard errors. Any absolute values greater than 2 are significant at the .05 level (Snijders et al., 2013).

p* < .01, *p* < .05; ****p* < .001.

RQ3 examines if serial participants' individual characteristics (organizational vs. individual accounts and type of account) significantly influence the structure of the serial participants' network over time. The analysis revealed strong social selection effects. Specifically, in terms of the difference between individual (coded as 1) vs. organizational (coded as 2) serial participants, the significant and positive alter effect suggests that in interactions, organizations prefer to make ties with other organizations. Individuals are less preferred in interactions. When combined together with the similarity effect (which indicates that serial participants are far more likely to interact with others of the same category), these results suggest that organizations prefer to interact with other organizations. Although individuals also prefer to interact with organizations, their ties are not reciprocated and thus they end up interacting more with other individuals.

The same kind of social selection is less apparent in terms of different types of serial participants. We found no significant alter or ego effects. It does not seem that being as any specific type of actor (e.g., news media versus advocacy organization) makes a difference in serial participants' level of popularity over time. Nonetheless, the negative and significant similarity effect suggests that serial participants do not prefer to interact with others from the same category.

RQ4 explores how serial participants' network positions in their CoP affect their network positions in the general public networks (this is the large network of all people who tweeted about climate during each event) over five years. To answer this question, we first calculated serial participants' network position in the serial participants' CoP and then their position in the overall networks of climate discussion in each event. We then examined the relationship between serial participants' network position in the participant community and in the general Twitter network. We ran the analysis on two types of centralities: eigenvector centrality, which reflects actors' connections with other important actors, and closeness centrality, which reflects actors' chance to act as brokers in networks.

We then used mixed-effects regression analysis (See Table 4). Figure 2 visualizes the overall trends for serial participants' network position changes and shows that they are gaining importance in the overall network over time. In order to control for the potential influence of serial participants' social media status (i.e., number of followers and followees) and characteristics of serial participant accounts (actor types and individual vs. organizational accounts), these variables and their interaction effect with time were also included in the models. Specifically, for eigenvector centrality, the analysis found a strong and persistent influence of serial participants' network positions in the participants' CoP on their position in the general network. This means that connections with elites in the serial participants' CoP help them obtain connections with elites in the general public's network. It is interesting to note that time alone failed to predict general network position. But the interaction effect of serial participants' eigenvector position and time significantly predicted change. This suggests that serial participants' positions in the general network did not change naturally as a function of time. The observed change in the general public network is a function of changes in serial participants' community positions. Moreover, another variable that significantly influenced changes in serial participants' network positions in the general network is the account's number of followers. The effect size of the number of followers ($r = .18$) is much smaller than that of serial participants' eigenvector centrality in the participants' community ($r = .47$). In other words, for serial participants to

Table 4. Results of the mixed-effect regression models with eigenvector centrality and closeness centrality as dependent variables.

	Eigenvector centrality						Closeness centrality					
	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE
Intercept	.001	.002	.000	.000	-.000	.006	.181***	.021	.190***	.022	.135	.042
Time	.000	.000	.000	.000	.000	.001	.003	.006	-.004	.007	-.002	.007
Serial Activist Network Position	.300***	.016	.288***	.016	.284***	.016	.137**	.046	.128**	.045	.140**	.046
Network Position*Time	-.031***	.004	-.030***	.005	-.030	.005	-.000	.013	-.001	.012	-.002	.007
Follower	—	—	.003***	.000	.000	.000	—	—	.000	.000	.000	.000
Follower*Time	—	—	.001***	.000	.000	.000	—	—	.000	.000	.000	.000
Friend	—	—	.000	.000	-.000	.000	—	—	-.000	.000	-.000	.000
Friend*Time	—	—	.000	.000	.000	.000	—	—	.000	.000	.000	.000
Organization/individual	—	—	—	—	.005	.002	—	—	—	—	.033**	.015
Actor Type	—	—	—	—	-.000	.000	—	—	—	—	.000	.016

* $p < .01$; ** $p < .05$; *** $p < .001$.

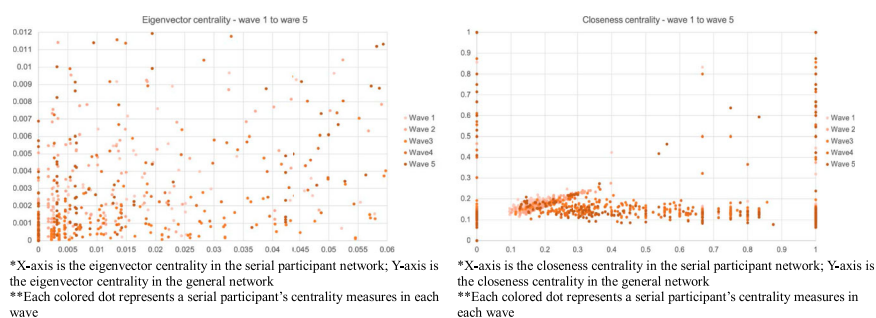


Figure 2. The changing dynamics of serial activists' eigenvector centrality and closeness centrality.

raise their eigenvector centrality in the general discussion network, they need to connect with other important serial participants and grow their follower base.

The second set of models examined the effect of serial participants' closeness centrality in the participant community. Our purpose was to see if brokerage roles in the serial users' community allow participants to increase their chance to act as brokers in the general public's network. The analysis again found that high closeness centrality in the participant community is strongly and positively associated with having high closeness centralities in the general public network. In addition, the analysis found that status as individual vs organization also influences changes of closeness centrality over time. Being an organization, over time, gives a serial participant more opportunity to act as a relationship broker in the larger network of people discussing climate on Twitter.

In short, although the size of serial participants' CoP is very small in comparison to the numbers of accounts that tweet about climate more sporadically, participants' position in this CoP matters. Enjoying a central, broker-like position in the serial participant community allows serial users to emerge as central and influential in the larger network.

Discussion

This study set out to examine the structural dynamics of a serial participants' community, and the value of such communities in the context of social media discourse related to climate issues. A number of findings emerge from the analysis that warrant further discussion.

Social selection sustains serial participants' community of practice

Communities are critical for sustained involvement in activism (Freelon et al., 2018). Although activists' communities in a single event or location have been extensively studied (Sobré-Denton, 2016), much less is known about whether activists who simply participate in digital discourse on an issue, across events, also interact like a community of practice (CoP). Moreover, although previous research has found the emergence of networked communities on social media based CoPs (Gilbert, 2016) and has documented different structural roles community members take on in CoPs (Weiss & Domingo, 2010; Wenger et al., 2002), little research has been done to provide explanation of how roles are assigned in informal settings, if hierarchies exist among geographically distant members, and which actors emerge as important in the process.

Findings of the current study suggest that the process of social selection generates micro structures that hold communities together (Block, 2015; Rambaran et al., 2015). We modeled several empirical indicators of social selection and our findings support the existence of a tightly knitted community of practice of Twitter users who are attentive to the climate change issue. First, the negative outdegree centrality effect suggested that actors tended to be selective in their choice of contacts. Second, we found positive reciprocity among serial participants. Reciprocity is widely considered a prerequisite of friendship. Their tendency to reciprocate ties suggest that they recognize each other on social media and formed long-term relationships with specific others. This is remarkable behavior considering that the overall climate discussion network involves half a million users who are geographically sparse.

Similarly, we found a positive transitive effect. In natural interaction settings, people who have a common friend are likely to meet and spend time together and thus become friends. We observe a similar effect in the serial participant community, suggesting the emergence of dyads and triads that are the building blocks of communities. Third, and most importantly, we found a negative three-cycle effect, which reveals a hierarchical interaction pattern. This finding can be explained by a theoretical model proposed by Davis and Leinhardt (1970), who argued that an unreciprocated tie in a cyclic configuration indicates a hierarchy difference. In other words, among three actors, one recipient of a tie occupies a higher position than the sender. The question to further explore, then, is who occupies the advantageous position? We explore this question in the next section.

The structural interaction patterns revealed in the analysis are highly similar to longitudinal interaction patterns commonly found in friendship and learning networks (Block, 2015; Rambaran et al., 2015). These findings support the idea that serial participants' issue communities are a form of CoP. Previous research on social mediated CoPs have generally found the emergence of networked communication (Gilbert, 2016; Zappavigna, 2012). The current study contributes to this body of literature by suggesting that networked social selection should be an important mechanism in social mediated CoP theorization. Future studies may develop theories to explain conditions in which different social selections occur and how social selections shape CoPs.

Factors driving social selections among serial participants

Previous social mediated CoP studies tend to assume that because social media offer open interaction platforms, social mediated CoPs should also have flat network structures (Stephansen & Couldry, 2014). In the current study we turn attention to social selections, which naturally produce social hierarchies (Lewis et al., 2012). Our analysis reveals that serial participants are aware of each other's similarities and differences in terms of organization vs. individual status and participant types. Moreover, these characteristics influence serial participants' choice of contact. We found that organizational actors emerged as the preferred interaction contacts over time. Our findings therefore challenge scholars to reconsider assumptions about the issue of hierarchies in social mediated CoPs.

The social movement literature has abundant discussion about the difference between organizational and individual participants (Diani & McAdam, 2003). In recent years, many scholars have argued that in the Web 2.0 era, the power of organizations are fading

away and individual participants have gained power and influence (Bennett & Segerberg, 2012). To a degree, our study confirms this observation. Among top serial participants in our data, individual participants play a consistent and important role.

However, as alluded to in the previous section, there seems to be a hierarchical local structure in the serial participant network. In many events, organizations are likely to be active top serial participants. What makes them even more important than individual participants is their role in the long-term evolution of serial participant communities: organizations are the preferred contacts in social interactions. At the same time, they are less likely to reciprocate such relationships with individuals. Instead, organizational serial participants are more likely to interact with other organizations. It is also interesting to note that the specific types of organizations (i.e., media organization, activist group) did not significantly influence tie formation pattern. Taken together, one possible explanation for these findings is that association with organizations of any kind indicates a higher level of status, legitimacy, and/or authority in the serial participant network. While individual participants seek to obtain such status, organizational participants are more likely to strategically use their connections with others with similar status. Our analysis suggests that in the serial participant community, organizations possess structurally important positions that foster relationship formation and their importance may be reinforced over time through communicative behaviors. It is also worthwhile to note that the context of climate change may also influence organizations' tie formation patterns. As a well-established social issue, many organizations devote considerable resources to address climate change and engage publics (Hestres, 2014, 2015). Future studies should examine whether organizations' roles change in movements related to emerging or more controversial social issues.

The value of serial participant CoP

Our analyses revealed a consistent, significant, and positive impact of serial participants' CoP positions on their position in the broader network of Twitter users discussing the climate issue. Based on analyses of eigenvector centralities, we found that users' elite connections in the serial participants' community significantly increased their chances to connect with elites in the general public issue network. The finding suggests when a serial participant connects with other elite serial participants in the serial participant community, they are also more likely to build connections with elites in the broader issue network. One explanation is that elite serial participants are widely regarded as leaders or authorities on the climate change issue. Building connections with them also raises serial participants' visibility and connection to elites outside their circle (Caniglia, 2001; Yoo & Lee, 2009). Komorowski et al.'s (2018) research on CoPs on Twitter also found that some community members (that also tend to be organizations) hold more prominent network positions than others and become the center of interactions. Our study extends previous research and suggests that one benefit of interacting with elites in CoPs is to raise members' network centrality within specific CoPs and beyond.

Additionally, we examined serial participants' closeness centrality, which reflects their likelihood to connect with diverse others. We found that, again, diverse connections in the serial participants' community increase participants' chance to obtain diverse ties in the general public network of Twitter discussions about climate. This finding suggests that when serial participants connect with other serial participants who have diverse

connections, it greatly increases their chances to obtain diverse connections within the broader discussion network as well. Overall, the analysis showed that the serial participant CoP offered critical connections to elites and enhanced actors' connections with diverse others. These are important values that serial participants can obtain from their membership in serial participant communities.

Conclusion

We demonstrated that serial participants can form a community of practice (CoP) via sustained involvement in social media discourse on climate issues across a diverse set of events. Moreover, the study revealed that serial participants' CoP offered them opportunities to connect with elites and function as brokers in the general public's network. Future studies may further examine other values offered by such communities and power dynamics that may facilitate or hinder the sustainability of these communities. Last but not least, future studies with larger sample size may consider how the presence of counter-movements may influence the dynamics in serial participants' CoPs.

Notes

1. Reciprocation describes the increased likelihood of individuals to send ties to those from whom they receive a tie. Transitivity of a relation means that when there is a tie from i to j , and also from j to h , then there is also a tie from i to h . The three-cycle refers to a triad of actors i , j , and h , it describes the configuration where a tie from i to j , a tie from j to h , and a tie from h to i is present.
2. Detailed coding definitions and codebook available from authors upon request.

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